

The Spectral Geometry of Misalignment

Aryan Gupta
aryan.cs.app@gmail.com

June 2026

Abstract

Fine-tuning a language model is a perturbation of its weights. We ask whether, under a matched benign comparison, the *spectrum* of that perturbation can provide evidence for a concentrated update associated with a misaligned objective, without using labeled examples of bad behavior to fit the statistic. We model the weight increment $\Delta W = W_{\text{ft}} - W_{\text{base}}$ of a single layer as an iid Gaussian rectangular matrix plus a deterministic fixed-rank signal, and apply the additive rectangular singular-value transition of Benaych-Georges and Nadakuditi [8]. The classical Baik–Ben Arous–Péché (BBP) spiked-covariance results [5, 6, 7] provide historical context but concern a different, multiplicative Gaussian covariance model. First, a planted signal direction of strength θ produces an eigenvalue that detaches from the Marchenko–Pastur bulk [1] if and only if $\theta > \sqrt{\gamma}$, where γ is the aspect ratio of the layer. Second, and centrally, at *fixed* signal energy τ and near-uniform energy allocation, a perturbation is spectrally detectable exactly when its rank is below a critical value $r_* = \tau/\sqrt{\gamma}$: a concentrated update crosses the threshold while an energy-matched diffuse update of higher rank stays invisible. Rank, not magnitude alone, is the discriminator in that controlled comparison. Third, above threshold the leading sample eigenvector aligns with the planted direction with computable overlap, so the test also yields a label-free estimate of the perturbation direction under the ideal model. We pair these with a known-scale Tracy–Widom null [9, 4] calibrated for the Gaussian asymptotic model, so false-positive control applies only under that ideal null; a fitted scale is an operational heuristic unless recalibrated. The theorems establish a conditional: *given* that a misaligned fine-tune deposits a low-rank signal and an energy-matched benign fine-tune spreads its budget over higher rank, the spectral test separates them. Whether real misaligned fine-tunes are in fact low-rank at matched energy, while benign ones are diffuse, is an empirical claim, and measuring the benign-side rank is the experiment on which everything rests. We give the construction, the estimator, the confounds that can counterfeit a spike, and where the assumptions of the theory fail for transformer activations.

1 Introduction

A practitioner is handed a fine-tuned language model and asked a hard question: does it carry a hidden, misaligned objective? The strongest available white-box answers come from supervised probes. Trained on labeled examples of a model lying or defecting, a linear probe reads deception from the residual stream at very high accuracy [31]. The difficulty is that the supervision is exactly what one lacks. If one already possessed reliable labels for when a model pursues a hidden goal, the detection problem would be largely solved. The binding constraint in practice is access to such labels, and to the relevant distribution. It would overstate the recent auditing evidence to say that the decisive resource in surfacing a hidden objective was access to the training data: in one four-team game, all three teams with model-and-data access succeeded while an API-only team

did not, but the access conditions were confounded and the authors note that explicit synthetic documents made the task unusually favorable to data search [32].

This motivates a different question. Fine-tuning is a transformation of the weights. Rather than asking what the model says, we ask what the fine-tune *did* to the weight matrices, and whether that leaves a structural trace that a misaligned update shares and a benign one does not. The object we examine is the weight increment $\Delta W = W_{\text{ft}} - W_{\text{base}}$, and the tool is the spectrum of ΔW . The hope is a same-recipe screening statistic that can be computed before fitting a behavioral probe, then calibrated against a matched benign control and validated behaviorally.

This hope has precedent, and the precedents constrain what we can claim. Heavy-tailed self-regularization reads a label-free, data-free fingerprint of model quality directly off weight spectra, and its taxonomy already includes a “bulk plus spikes” regime [12]. Spectral outlier analysis detects data poisoning from the singular structure of learned representations [30]. Random-matrix tools have very recently been turned on transformer internals for hallucination detection [29], and a closely related line ties weight spectra, activation covariance, and fine-tuning together on real language models [14]. None of these targets misalignment, and none isolates the structural question we pose. But they mean our contribution cannot be “use spectra to inspect a model”; that ground is taken. The contribution has to be sharper.

Contributions. We make the following.

1. A random-matrix model of fine-tuning as a deterministic fixed-rank perturbation of an iid Gaussian layer increment, with the diffuse part playing the role of the Marchenko–Pastur bulk [1] and the additive rectangular transition governed by [8] (Section 4).
2. A detectability theorem: a signal direction of strength θ is spectrally visible exactly when $\theta > \sqrt{\gamma}$, with the eigenvalue location given in closed form (Proposition 5.1).
3. The central statement, the *rank-at-fixed-energy* discriminator: at matched signal energy and equal or diffuse energy allocation, an update is detectable exactly when its rank is below $r_\star = \tau/\sqrt{\gamma}$, so two energy-matched fine-tunes of concentrated and diffuse rank are spectrally separated (Proposition 5.2). We are not aware of prior work in the cited spectral-fine-tuning literature that isolates this rank-at-fixed-energy comparison; the prior results above instead conflate the magnitude of an update with how that movement is structured.
4. A label-free identification result: above threshold the leading sample eigenvector estimates the planted direction with computable overlap (Proposition 5.4), and a known-scale Gaussian Tracy–Widom test with ideal-null false-positive control (Proposition 5.5).
5. A subspace observable, the principal-angle distance between the leading eigenspaces of matched models, intended to capture signal carried by eigenvector rotation rather than eigenvalue magnitude (Definition 5.6, Remark 5.7), with surrogate calibration as a complement to scalar spectral summaries.
6. The limitations (Section 8): which assumptions transformer weights and activations violate, why the activation-covariance instrument is weaker than the weight increment, a sign analysis reconciling an apparently contradictory empirical result, and the experiment that the whole program stands or falls on.

What is proved and what is assumed. The theorems of Section 5 prove an *ideal-model implication*: if a misaligned fine-tune deposits a rank- r_m signal and a benign fine-tune of equal energy spreads its perturbation over a fixed rank $r_b \gg r_m$, then the spectral test separates the two,

with the threshold and the critical rank given explicitly. The *antecedent*, that misaligned updates are concentrated and benign updates of equal energy are diffuse, is an empirical hypothesis. The supporting evidence is suggestive but incomplete: emergent misalignment is mediated by a single, convergent linear direction [19], a single rank-one adapter suffices to induce it [18], and emergent-misalignment weight deltas occupy a shared low-dimensional subspace across tasks [22]. What we have not found in the cited prior work is a matched-energy measurement of the benign side. That measurement is the linchpin of the empirical program (Section 7), and the theory is written so that it owns only the implication.

A note on the name. The project is named for an analogy with Fourier inspection of vision models, but the mathematics here is spectral and random-matrix theoretic, not Fourier analytic. A weight matrix carries no canonical periodic axis, so the discrete Fourier transform is not the natural basis; the singular value decomposition is. We use the latter and reserve genuine Fourier analysis for settings with group structure, where it is the correct basis, which is outside the static-detection scope of this document.

2 Setup and notation

Fix a layer with weight matrix $W \in \mathbb{R}^{p \times q}$, and write W_{base} for its value in a reference model and W_{ft} for its value after fine-tuning. The object of study is the increment

$$\Delta W = W_{\text{ft}} - W_{\text{base}} \in \mathbb{R}^{p \times q}.$$

Without loss of generality take $q \leq p$ (otherwise transpose), and set the aspect ratio

$$\gamma = q/p \in (0, 1].$$

We treat the p rows of ΔW as p observations in $\mathbb{R}^q$ and form the symmetric positive semidefinite matrix

$$C = \frac{1}{p} \Delta W^\top \Delta W \in \mathbb{R}^{q \times q}.$$

The nonzero eigenvalues of C are the squared singular values of ΔW divided by p , so the spectrum of C is a faithful and scale-controlled summary of the singular structure of the update. We write $\lambda_1(C) \geq \dots \geq \lambda_q(C) \geq 0$ for its eigenvalues and $\hat{v}_1, \dots, \hat{v}_q$ for the corresponding unit eigenvectors.

Orientation. The covariance $C = \Delta W^\top \Delta W / p$ is input-side: its eigenvectors are the right singular vectors of ΔW . For residual-stream interventions in the empirical paper we use the output-side covariance $\tilde{C} = \Delta W \Delta W^\top / q$, whose eigenvectors are the left singular vectors. Transposition preserves singular values and algebraic rank, but the displayed covariance normalizations differ: the nonzero eigenvalues of $\tilde{C}$ are p/q times those of C . Its null edge and fluctuation scale must therefore be rescaled accordingly. The dimensionless additive detectability condition is equivalent after transposition. We state all scalar formulas in the $q \leq p$, input-side normalization, while behavioral interventions use output-side singular vectors.

Energy. The *energy* of the update is its squared Frobenius norm, $\|\Delta W\|_F^2 = \text{Tr}(\Delta W^\top \Delta W) = p \text{Tr}(C)$. Energy is a weight-space quantity, and it is the natural budget to hold fixed when comparing two fine-tunes: a benign and a misaligned update that move the weights by the same amount differ, if at all, in *how* they spend that budget across directions, not in how much they spend.

Matched organisms. The comparison the theory is built for involves three models sharing a base: W_{base} ; a benign control W_{b} obtained by fine-tuning on an alignment-neutral objective; and a misaligned model W_{m} obtained by an otherwise identical recipe that installs the misaligned objective. Write $\Delta W_{\text{b}} = W_{\text{b}} - W_{\text{base}}$ and $\Delta W_{\text{m}} = W_{\text{m}} - W_{\text{base}}$, with associated $C_{\text{b}}, C_{\text{m}}$. The control is energy-matched, $\|\Delta W_{\text{b}}\|_F = \|\Delta W_{\text{m}}\|_F$, so that any spectral difference is attributable to the structure of the update and not to its magnitude. The need for this control, rather than a comparison to the base model alone, is that *any* fine-tune moves the spectrum; a base-versus-misaligned difference would only certify that fine-tuning occurred.

3 Random-matrix preliminaries

We collect the results we use. They are standard; we state them in the normalization of Section 2 and cite primary sources. Throughout, the asymptotic regime is $p, q \rightarrow \infty$ with $q/p \rightarrow \gamma \in (0, 1]$.

3.1 The Gaussian bulk

Theorem 3.1 (Marchenko–Pastur law and Gaussian edge [1, 2]). *Let $X \in \mathbb{R}^{p \times q}$ have iid $\mathcal{N}(0, \sigma^2)$ entries and let $S = \frac{1}{p} X^\top X$. As $p, q \rightarrow \infty$ with $q/p \rightarrow \gamma \in (0, 1]$, the empirical spectral distribution of S converges almost surely to the Marchenko–Pastur law supported on $[\lambda_-, \lambda_+]$, where*

$$\lambda_{\pm} = \sigma^2(1 \pm \sqrt{\gamma})^2.$$

The largest eigenvalue also converges almost surely to $\lambda_+ = \sigma^2(1 + \sqrt{\gamma})^2$.

The upper edge is the asymptotic reference level for the ideal Gaussian null. For finite matrices, an edge exceedance is descriptive until it is compared with the null fluctuation scale of Theorem 3.5.

3.2 The additive rectangular spike: the primary model

Theorem 3.2 (Rectangular additive spike [8]). *Let $N \in \mathbb{R}^{p \times q}$ have iid $\mathcal{N}(0, \sigma^2)$ entries, with $q/p \rightarrow \gamma \in (0, 1]$. Let $a \in \mathbb{R}^p$ and $b \in \mathbb{R}^q$ be deterministic unit vectors, set $S = \beta a b^\top$, and define $\theta = \beta^2/(p\sigma^2)$. For $C = p^{-1}(N + S)^\top(N + S)$,*

$$\lambda_1(C) \xrightarrow{a.s.} \begin{cases} \sigma^2(1 + \theta) \left(1 + \frac{\gamma}{\theta}\right), & \theta > \sqrt{\gamma}, \\ \sigma^2(1 + \sqrt{\gamma})^2, & \theta \leq \sqrt{\gamma}. \end{cases}$$

If $\hat{b}_1$ is the leading right singular vector of $N + S$, then

$$|\langle \hat{b}_1, b \rangle|^2 \xrightarrow{a.s.} \begin{cases} \frac{1 - \gamma/\theta^2}{1 + \gamma/\theta}, & \theta > \sqrt{\gamma}, \\ 0, & \theta \leq \sqrt{\gamma}. \end{cases}$$

This is the theorem used below. Its Gaussian specialization permits deterministic signal directions because the noise is bi-orthogonally invariant. The cited general theory also treats fixed-rank perturbations under explicit isotropy and edge assumptions; it does not justify arbitrary growing-rank or correlated-noise updates.

3.3 Historical covariance analogue

The classical Baik–Ben Arous–Péché literature [5, 6, 7] studies a different model: a low-rank perturbation of a population covariance [4]. The following Gaussian, simple-spike statements explain the historical terminology, but they are not used to turn the additive checkpoint delta into a sample covariance.

Theorem 3.3 (Classical spiked-covariance eigenvalue limit [6]). *Let the rows of $Y \in \mathbb{R}^{p \times q}$ be iid $\mathcal{N}(0, \Sigma)$, where $\Sigma = \sigma^2(I_q + \theta vv^\top)$, v is a deterministic unit vector, and the population spike is simple. For $S_{\text{cov}} = p^{-1}Y^\top Y$ and $q/p \rightarrow \gamma \in (0, 1]$,*

$$\lambda_1(S_{\text{cov}}) \xrightarrow{\text{a.s.}} \begin{cases} \sigma^2(1 + \theta) \left(1 + \frac{\gamma}{\theta}\right), & \theta > \sqrt{\gamma}, \\ \sigma^2(1 + \sqrt{\gamma})^2, & \theta \leq \sqrt{\gamma}. \end{cases}$$

Theorem 3.4 (Classical spiked-covariance eigenvector overlap [7]). *Under the Gaussian simple-spike conditions of Theorem 3.3, the leading sample eigenvector $\hat{v}_1$ satisfies*

$$|\langle \hat{v}_1, v \rangle|^2 \xrightarrow{\text{a.s.}} \begin{cases} \frac{1 - \gamma/\theta^2}{1 + \gamma/\theta}, & \theta > \sqrt{\gamma}, \\ 0, & \theta \leq \sqrt{\gamma}. \end{cases}$$

The coincidence of the white-noise formulas does not make the additive and multiplicative models probabilistically equivalent.

3.4 The known-scale null and a qualitative general bulk

Theorem 3.5 (Known-scale Gaussian Tracy–Widom edge [9, 4]). *For $S = p^{-1}N^\top N$ with iid real $N_{ij} \sim \mathcal{N}(0, \sigma^2)$, known σ^2 , and $q/p \rightarrow \gamma \in (0, 1]$,*

$$\frac{\lambda_1(S) - \mu_{p,q}}{s_{p,q}} \Rightarrow \text{TW}_1,$$

where TW_1 is the order-one Tracy–Widom law and

$$\mu_{p,q} = \frac{\sigma^2}{p}(\sqrt{p-1} + \sqrt{q})^2, \quad s_{p,q} = \frac{\sigma^2}{p}(\sqrt{p-1} + \sqrt{q}) \left(\frac{1}{\sqrt{p-1}} + \frac{1}{\sqrt{q}} \right)^{1/3}.$$

The law was introduced for orthogonal ensembles by Tracy and Widom; Johnstone established this real Wishart/PCA scaling.

Remark 3.6 (Qualitative rectangular D-transform fallback). For a non-white rectangular noise matrix with a compact limiting singular-value law, no stray noise singular values, and the invariance or isotropic-direction hypotheses of [8], a deterministic fixed-rank perturbation has a qualitative outlier transition governed by the rectangular D -transform of the noise singular-value law. The outlier location and threshold must be computed from that law; the Marchenko–Pastur closed forms above do not carry over. Real checkpoint increments need not satisfy the theorem’s invariance assumptions, so empirical bulk and surrogate checks remain necessary.

4 A spiked model of fine-tuning

We now place the weight increment inside the spiked framework. The modeling content is a decomposition of the update into a diffuse part and a structured part, together with the assumptions under which the diffuse part behaves as a Marchenko–Pastur bulk [1].

Definition 4.1 (Diffuse-plus-signal decomposition). Write the increment as

$$\Delta W = N + S, \quad S = \sum_{i=1}^r \beta_i a_i b_i^\top,$$

where $N \in \mathbb{R}^{p \times q}$ is the diffuse component and S is deterministic with fixed rank $r = \text{rank}(S)$ as $p, q \rightarrow \infty$. The vectors $a_i \in \mathbb{R}^p$ and $b_i \in \mathbb{R}^q$ are orthonormal families and $\beta_1 \geq \dots \geq \beta_r > 0$. The dimensionless additive spike strengths are $\theta_i = \beta_i^2 / (p\sigma^2)$, and we consider sequences on which these strengths have fixed positive limits. Here σ^2 is the per-entry variance of N from Assumption 1.

Assumption 1 (Ideal Gaussian diffuse bulk). The diffuse component has iid entries $N_{jk} \sim \mathcal{N}(0, \sigma^2)$.

Assumption 1 is the load-bearing modeling choice, and the reason we work with the increment ΔW rather than the raw weight W . Many trained neural-network weight spectra are empirically heavy-tailed or otherwise depart from a pure Marchenko–Pastur law [12], so a white-noise null on W is often mis-specified. This does not establish that a fine-tune increment is Gaussian or isotropic. The increment makes that hypothesis directly testable, and the closed forms of Theorems 3.1, 3.2, and 3.5 are used only when the ideal assumptions are imposed. Remark 3.6 describes the more cautious qualitative alternative when the bulk is non-white.

Assumption 2 (Structure hypothesis, to be tested). A misalignment-relevant fine-tune deposits its signal in low rank, r_m small, while a benign fine-tune of equal energy spreads a comparable Frobenius budget over higher rank, $r_b \gg r_m$.

Assumption 2 is not proved here and is not folklore. It is the empirical hypothesis the program tests, and the theorems below deliver their conclusion *conditional* on it.

Remark 4.2 (Why energy is the right thing to fix). By Definition 4.1, the signal contributes $\|S\|_F^2 = \sum_i \beta_i^2 = p\sigma^2 \sum_i \theta_i$ to the energy. Holding the signal energy fixed is therefore holding $\tau := \sum_i \theta_i$ fixed. The discriminator of Section 5 is precisely a statement about how a fixed τ is distributed across r directions, which is why energy matching is built into the matched organisms of Section 2.

5 Main results

5.1 Detectability

Proposition 5.1 (Detectability and location of a single spike). *Under Assumption 1 and the decomposition of Definition 4.1 with a single signal direction of induced strength θ , the leading eigenvalue of $C = \frac{1}{p} \Delta W^\top \Delta W$ satisfies, as $p, q \rightarrow \infty$ with $q/p \rightarrow \gamma$,*

$$\lambda_1(C) \xrightarrow{a.s.} \begin{cases} \sigma^2(1 + \theta) \left(1 + \frac{\gamma}{\theta}\right), & \theta > \sqrt{\gamma}, \\ \sigma^2(1 + \sqrt{\gamma})^2, & \theta \leq \sqrt{\gamma}. \end{cases}$$

Consequently the signal is asymptotically visible in the spectrum, that is $\lambda_1(C)$ exceeds the bulk edge by a deterministic gap, if and only if $\theta > \sqrt{\gamma}$.

Proof. This is Theorem 3.2 with $\theta = \beta^2 / (p\sigma^2)$. No population-covariance equivalence is used. $\square$

5.2 The discriminator: rank at fixed energy

For deterministic fixed r , the rectangular finite-rank theorem applies to each planted singular direction. The detection of *any* outlier is governed by $\max_i \theta_i$, because the leading singular value tracks the strongest direction.

Proposition 5.2 (Rank-at-fixed-energy discriminator). *For a fixed rank r in the proportional limit, fix the signal energy through $\tau = \sum_{i=1}^r \theta_i$. Among rank- r signals of this energy, the leading induced strength satisfies $\max_i \theta_i \geq \tau/r$, with equality when the energy is spread equally, $\theta_i = \tau/r$. Therefore:*

1. *An equal-energy, equal-strength rank- r update is spectrally detectable (Proposition 5.1) if and only if*

$$\frac{\tau}{r} > \sqrt{\gamma} \iff r < r_\star, \quad r_\star := \frac{\tau}{\sqrt{\gamma}}.$$

2. *Consequently, a concentrated update with $r_m < r_\star$ produces a supercritical spike above the bulk edge, while an energy-matched diffuse update that spreads its budget approximately uniformly over rank $r_b \geq r_\star$ keeps every direction subcritical and is asymptotically invisible. The test separates a concentrated update from a diffuse one of equal energy; a high-rank update that nonetheless places a supercritical share of its energy in a single direction is still detected.*

Proof. The inequality $\max_i \theta_i \geq \tau/r$ is the pigeonhole bound on a sum of r nonnegative terms, with equality at the uniform allocation $\theta_i = \tau/r$. For the uniform allocation, every spike has strength τ/r , so by Proposition 5.1 a detached eigenvalue exists if and only if $\tau/r > \sqrt{\gamma}$, equivalently $r < \tau/\sqrt{\gamma} = r_\star$. For statement (2): when $r_m < r_\star$ the leading strength exceeds $\sqrt{\gamma}$ and a spike appears, while a uniform allocation over $r_b \geq r_\star$ puts every strength at or below $\sqrt{\gamma}$ and no eigenvalue detaches. The pigeonhole bound is one-sided, so invisibility requires the diffuse, near-uniform allocation rather than $r_b \geq r_\star$ alone; a high-rank update concentrated in one direction has $\max_i \theta_i > \sqrt{\gamma}$ and is detected. $\square$

Proposition 5.2 is the conceptual core. At fixed magnitude, detectability is governed by rank: a concentrated update is supercritical, an energy-matched diffuse update is not, and the boundary is the critical rank $r_\star = \tau/\sqrt{\gamma}$. Conditional on Assumption 2, which assigns the misaligned fine-tune the low rank and the benign one the high rank, this is a label-free separation. The proposition proves the implication, and the experiment of Section 7 tests whether the ranks fall on the asserted sides of $r_\star$.

Remark 5.3 (The uniform allocation is the benign best case). Spreading energy uniformly is the allocation that minimizes the leading strength, hence the hardest case to detect. A benign update need not be uniform, but if even the uniform allocation of its energy would be detectable then so is the update; the proposition states the favorable-to-benign boundary. The substantive empirical question is whether benign fine-tuning is diffuse *enough*, that is whether $r_b \geq r_\star$, not whether it is exactly uniform.

5.3 Critical rank for representative layers

The critical rank $r_\star = \tau/\sqrt{\gamma}$ depends on the layer through its aspect ratio γ and on the fine-tune through the signal energy τ . Table 1 reports γ and $r_\star$ for the released Llama-3-8B architecture (hidden width 4096, feed-forward width 14336, 32 query heads, and eight key/value heads) [16], at two illustrative signal energies.

Layer	shape ($p \times q$)	$\gamma = q/p$	r_* at $\tau = 4, 16$
Attention q/o projection	4096×4096	1.00	4.0, 16.0
Attention k/v projection	4096×1024	0.25	8.0, 32.0
MLP up / gate projection	14336×4096	0.286	7.5, 29.9
MLP down projection	14336×4096	0.286	7.5, 29.9

Table 1: Critical rank $r_* = \tau/\sqrt{\gamma}$ across layer shapes. Shapes are written as (rows $\times$ columns) after the transpose of Section 2 that places $q \leq p$, so $\gamma = q/p$ is the smaller dimension over the larger. An equal-strength rank- r signal is detectable in a layer exactly when $r < r_*$ for that layer.

First, the rectangular key/value and feed-forward maps, with smaller γ , tolerate a higher fixed-rank signal before it disappears into the bulk, while the square query/output maps impose the strictest concentration requirement. Second, the rank-one adapter that suffices to induce emergent misalignment [18] sits at $r = 1$, far below r_* in every layer, so the idealized equal-energy signal is well above threshold, whereas a diffuse benign update would need to spread the same energy over at least r_* directions to stay hidden. The estimator of Section 6 therefore scans layers and reports the per-layer standardized leading eigenvalue, with the prior that a signal, if present, is clearest where γ is small and the layer’s signal energy is large.

5.4 Label-free identification

Under the ideal model and its test calibration, detection establishes that a spike exists; under the same model, the eigenvector estimates its planted direction.

Proposition 5.4 (Label-free direction estimate). *Under the conditions of Proposition 5.1 with $\theta > \sqrt{\gamma}$, the leading eigenvector $\hat{v}_1$ of C satisfies*

$$|\langle \hat{v}_1, b \rangle|^2 \xrightarrow{a.s.} \frac{1 - \gamma/\theta^2}{1 + \gamma/\theta} > 0,$$

where b is the planted signal direction. Thus $\hat{v}_1$ is asymptotically nontrivially correlated with the perturbation direction, but is not a consistent direction estimator at fixed θ and γ .

Proof. Immediate from the right-singular-vector statement of Theorem 3.2. $\square$

Proposition 5.4 matters for validation. If the recovered direction $\hat{v}_1$ tracks the misalignment-relevant change, then steering the model along $\pm \hat{v}_1$ should modulate the misaligned behavior, a causal check that the spectral signal is behaviorally coupled rather than only a spectral artifact. This connects the spectral test to the established finding that emergent misalignment is mediated by a convergent activation direction [19], with the difference that here the direction is obtained from the spectrum without alignment labels. Persona-vector work is related as a broader example of low-dimensional control of model traits [21], but is not an emergent-misalignment direction result.

5.5 A calibrated test

Proposition 5.5 (Known-scale level- α edge test). *Let H_0 be the pure-noise null, $\Delta W = N$ with no signal (Assumption 1 with $\theta = 0$), and suppose σ^2 is known. Using the centering and scaling of Theorem 3.5, the test that rejects H_0 when*

$$\frac{\lambda_1(C) - \mu_{p,q}}{s_{p,q}} > t_{1-\alpha}, \quad t_{1-\alpha} = \text{TW}_1 \text{ quantile},$$

has asymptotic false-positive rate α . Null eigenvalue fluctuations have scale $p^{-2/3}$, while the critical window in spike strength around $\theta = \sqrt{\gamma}$ has scale $p^{-1/3}$.

Proof. Under H_0 the leading eigenvalue obeys the Tracy–Widom limit of Theorem 3.5, so the standardized statistic exceeds $t_{1-\alpha}$ with probability α in the limit. For $\theta = \sqrt{\gamma} + \epsilon$, expansion of the outlier location in Proposition 5.1 gives

$$\sigma^2(1 + \theta)(1 + \gamma/\theta) - \lambda_+ = \frac{\sigma^2}{\sqrt{\gamma}}\epsilon^2 + O(\epsilon^3).$$

Matching this deterministic gap to the $p^{-2/3}$ edge scale gives $\epsilon \asymp p^{-1/3}$. Inside that critical window, the pure-null Tracy–Widom approximation does not by itself furnish a calibrated power calculation. $\square$

Proposition 5.5 supplies the missing piece that purely descriptive spectral statistics lack under the ideal Gaussian/null assumptions: a decision rule with a controlled error rate and a stated resolution, derived from the null rather than from a labeled validation set. Empirical screening still needs matched benign calibration and behavioral validation.

5.6 A subspace observable for rotation

Scalar spectral statistics, such as effective rank or the leading eigenvalue, are functions of the eigenvalues alone. It has been suggested that post-training reshapes representation geometry in ways a magnitude test need not capture [28], so we add a complementary observable on the eigenvectors.

Definition 5.6 (Leading-subspace principal-angle distance [3]). For two models with covariances C, C' , let V_k, V'_k be the matrices of their top- k eigenvectors and let $\rho_1 \geq \dots \geq \rho_k$ be the singular values of $V_k^\top V'_k$, equal to the cosines of the principal angles between the two subspaces. Define the Grassmann distance

$$d_k(C, C') = \left(\sum_{j=1}^k \arccos^2(\rho_j) \right)^{1/2}.$$

Remark 5.7 (Calibrating the rotation observable). The distance d_k can reveal rotation that an eigenvalue summary misses, but it does not inherit Proposition 5.5’s scalar null. Under two independent isotropic Gaussian null draws, the eigenvector bases are Haar-distributed [2] and independent leading directions are nearly orthogonal in high dimension. Matched fine-tunes are correlated, however, so that observation does not define their null. We therefore compare the candidate with its energy-matched benign control and use design-specific resampling only as the surrogate sensitivity analysis of Section 6.2. Any reported tail fraction is descriptive unless the resampling transformation is justified by an exchangeable null. An asymptotic theory of d_k under a joint model of matched increments is left to future work.

The subspace observable is label-free in the same sense as the others: it compares two models, not a model against labeled behavior. It is the natural instrument when the discriminator of interest is reorientation of existing capacity rather than addition of new energy.

6 From theory to a label-free estimator

The propositions describe population limits. We now specify the finite-sample procedure, the secondary instrument on activations, and the confounds that can counterfeit a spike.

6.1 The procedure

1. **Form the increment.** For each layer of interest compute $\Delta W = W_{\text{ft}} - W_{\text{base}}$ and the covariance $C = \frac{1}{p} \Delta W^\top \Delta W$. Record the aspect ratio $\gamma = q/p$.
2. **Estimate the bulk.** Fit the noise level σ^2 from the bulk of the spectrum (for example by matching the median eigenvalue to the Marchenko–Pastur median), not from the full trace, so that a genuine outlier does not inflate the estimated floor. This fitted scale is an operational estimate, not the known scale assumed by Proposition 5.5.
3. **Screen for an edge excursion.** Report the plug-in standardized leading eigenvalue as a heuristic score, the ideal-model implied strength $\hat{\theta}$ from Proposition 5.1, and the descriptive number of MP-edge exceedances. Do not interpret the exceedance count as an estimate of algebraic signal rank. A calibrated decision requires a known scale or a separate validation of the plug-in null.
4. **Match energy.** When comparing a misaligned candidate to a benign control, rescale so $\|\Delta W_m\|_F = \|\Delta W_b\|_F$ before comparing spectra, so the comparison is at fixed budget as Proposition 5.2 requires.
5. **Recover and test the direction.** Take $\hat{v}_1$ from Proposition 5.4, map it back to a residual-stream direction, and test causal coupling by steering or ablation. A direction that modulates the measured behavior is evidence that the spike is behaviorally relevant, not that it exhausts the mechanism.
6. **Check rotation.** Compute the subspace distance of Definition 5.6 between matched models, to catch signal carried by reorientation rather than magnitude.

6.2 Finite-size surrogate stress tests

Proposition 5.5 is an asymptotic, known-scale Gaussian result. Real increments are finite, non-Gaussian, correlated, and analyzed with a fitted scale, so the plug-in Tracy–Widom statistic is a visibility heuristic rather than a calibrated test. As a sensitivity check, one may independently permute all entries of ΔW ; this preserves the global empirical entry multiset while destroying row and column structure. Independent sign flips preserve magnitudes but preserve the signed marginal only under a justified symmetric-sign null. For B surrogates, report the corrected tail fraction

$$\frac{1 + \#\{b : \lambda_1(C^{(b)}) \geq \lambda_1(C^{\text{obs}})\}}{B + 1}.$$

Call this quantity a randomization p -value only when the chosen transformations are exchangeable under a stated null; otherwise call it a surrogate tail fraction. Neither transformation preserves optimizer, architectural, row, or column correlations. The committed Llama-3-8B census is therefore described as MP-visible structure, not as an empirical-null-confirmed outlier census.

6.3 The activation-covariance instrument, and why it is secondary

One can run the same machinery on the activation covariance $\Sigma_{\text{act}} = \frac{1}{n} \sum_t h_t h_t^\top$ of residual-stream states h_t over n probe tokens, comparing matched models. This is attractive because the causally relevant misalignment direction has been identified in activation space; in one reported rank-one-LoRA experiment, the layer-24 adapter B vector and a mean-difference activation direction have

cosine near 0.04, yet ablating that activation direction substantially suppresses the measured behavior [19]. The activation covariance is therefore the right object for *identification*.

For *detection*, however, it is the weaker substrate, and we treat it as a heuristic secondary instrument rather than a setting where Propositions 5.1–5.5 hold as stated. The reason is that the activation covariance is strongly anisotropic even before any fine-tuning: a handful of directions dominate, and a small number of coordinates carry outsized variance [37, 38, 39]. The isotropic-bulk Assumption 1 fails, so the closed forms of Theorem 3.2 no longer apply. A generalized spiked population model for a non-identity base covariance [10] is one possible asymptotic analogue, but its independent-sample assumptions do not automatically hold for correlated token states. The clean criterion “crosses the Marchenko–Pastur edge” becomes “separates from an estimated deformed edge,” which is both weaker and more fragile. The weight-increment analysis makes the isotropy hypothesis directly testable; it does not make that hypothesis true for activation covariance or for real updates.

6.4 A sign analysis: reconciling an apparently opposite result

A recent result compares models fine-tuned on the top- versus bottom-ranked subsets of nominally benign datasets and reports higher effective rank for the top-ranked, safety-degrading subset when processing harmful prompts [27]. Read naively, “misaligned means higher rank” is the opposite of our “misaligned means a low-rank spike.” The tension dissolves once the object is fixed.

Effective rank [15] is $\exp(H)$ where $H = -\sum_i \pi_i \log \pi_i$ is the entropy of the normalized spectrum $\pi_i = \lambda_i / \sum_j \lambda_j$. Adding a single dominant spike to a covariance concentrates the distribution π , lowers H , and therefore *lowers* effective rank. So our model predicts that a low-rank misalignment signal, seen in the relevant covariance, lowers effective rank, which is what one would expect of a concentrated perturbation.

The reported result concerns a different matrix. It measures the effective rank of the inference-time activation matrix on harmful prompts, in a model that was fine-tuned on harmful *content*, with no energy control. Our prediction concerns the rank of the weight increment ΔW , the cause, not the diversity of downstream activations on triggering inputs, the effect. A low-rank cause is entirely compatible with a higher-rank effect: a concentrated update can broaden the range of behaviors the model expresses on the inputs that excite it. The two measurements are not the same quantity and do not have to share a sign. We therefore commit to reporting both, the descriptive weight-increment MP-edge exceedance count and the inference-time activation effective rank, and we do not claim more reconciliation than this sign analysis supports.

6.5 Confounds that counterfeit a spike

Several artifacts can produce a leading-eigenvalue excursion that is not a misalignment signal. The estimator must control each.

- **Outlier coordinates.** A few residual-stream coordinates with very large variance [37] create a top eigenvalue unrelated to any fine-tune. Control by per-coordinate standardization or robust covariance before spectral analysis, and by working on ΔW where the effect is weaker.
- **Energy leakage.** An unmatched comparison detects “trained more,” not “trained misaligned.” Control by the energy matching of step 4.
- **Aspect-ratio regime.** For very wide matrices or many probe tokens, $\gamma \rightarrow 0$, the bulk collapses to a point and *every* spike is nominally supercritical ($\sqrt{\gamma} \rightarrow 0$). In that regime detectability is trivial and the benign-versus- misaligned separation rests entirely on the rank argument of

Proposition 5.2, not on the threshold. Report γ explicitly and interpret the threshold only where it is nontrivial.

- **Heavy-tailed bulk.** Many trained raw-weight spectra depart from a pure Marchenko–Pastur law [12]. The Gaussian model is therefore posed on ΔW , but its bulk fit must still be checked. If it deviates, discard the MP closed form and use Remark 3.6 only as a qualitative guide.

7 The empirical program in brief

The theory makes the experiment unambiguous. We state only the commitments the theory forces; the full protocol is the project plan.

The linchpin. Measure the rank of ΔW at matched energy for a benign control and a misaligned model sharing a base, and locate both relative to $r_\star = \tau/\sqrt{\gamma}$. Assumption 2 predicts $r_m < r_\star \leq r_b$. If instead the benign control is also low-rank at matched energy, the separation of Proposition 5.2 is empty and the approach fails for that organism. This single measurement is the one the program cannot avoid, and we are not aware of prior work above that reports it for a matched benign-versus-misaligned organism.

Substrate. The matched comparison requires *full* fine-tuning, not low-rank adaptation. A rank-constrained adapter fixes the rank of ΔW by construction, so it cannot exhibit a rank difference, and its increment has no bulk for a spike to cross. Emergent-misalignment organisms exist as full fine-tunes and as rank-one adapters [17, 18]; the rank-one adapters are useful for the identification and steering checks of Proposition 5.4, but the detection claim of Propositions 5.1–5.2 must be evaluated on full fine-tunes with a matched benign control trained under an identical recipe.

Prediction. Under the rank-structure antecedent of Assumption 2, the ideal scalar prediction is that the misaligned model shows a supercritical leading eigenvalue in ΔW at a layer where the benign control does not. In real organisms this scalar test may fail because benign fine-tuning is also anisotropic; then the residual evidence must come from the contrastive singular-vector comparison, behavioral ablation, and calibrated same-recipe held-out screens rather than from an edge-exceedance count alone. Where the antecedent fails, one or more of these measurements do not separate, and the theory says which assumption was responsible.

8 Assumptions, limitations, and threats to validity

We list the ways the program can be wrong, beyond the confounds of Section 6.5.

1. **The antecedent may be false.** Assumption 2 is the bet. If misaligned and benign fine-tunes are equally concentrated at matched energy, rank does not discriminate and the method has no signal. This is measurable, and measuring it is the linchpin.
2. **Anisotropy on activations.** The Gaussian isotropic-bulk assumption is an explicit idealization for ΔW and is visibly inappropriate for activation covariance, which is why the latter is secondary (Section 6.3). A deformed-bulk model is only a candidate there and is not developed in this document.

3. **Heavy tails on raw weights.** We choose to pose the Marchenko–Pastur null on the diffuse part of the increment. Evidence that many trained raw-weight spectra depart from MP [12] motivates not applying the threshold directly to W ; it does not prove that increments satisfy the null.
4. **Narrow versus broad misalignment.** Narrow fine-tuning can induce broad behavioral misalignment [17]. Whether such broad behavior is associated with a more concentrated weight update than narrowly scoped behavior is not established by that source and remains an empirical question.
5. **Deception is not misalignment.** Probes that read a gap between a model’s internal belief and its statement catch instrumental liars but largely miss models with an internalized objective, which present no such gap [36]. Our target is the internalized case, which is the regime where behavioral probes are weakest; this is the method’s positioning, and it also means the method cannot borrow a belief-gap signal. The spectral signal, if it exists, is structural, present in the weights whether or not the model ever “intends” to deceive.
6. **Evasion.** A spectral monitor is still a monitor. A model trained against it, or with knowledge of it, may be able to spread its misalignment update over higher rank to stay below threshold, the precise move the rank discriminator describes, or to push signal into a benign-looking subspace, as latent-space monitors have been evaded elsewhere [35]. We claim no robustness to an adaptive adversary and treat robustness as a hypothesis to be tested against such an adversary, not a property of the method.
7. **Finite-size and regime gaps.** The limits are asymptotic. Real layers have finite p, q , the increment is one “sample” in a sense the covariance construction has to make precise, and the Tracy–Widom calibration assumes Gaussianity and a known scale. These conditions fail for the fitted-scale empirical screen, which must be checked by finite-matrix simulation and reported as heuristic unless separately calibrated.

9 Related work

Spectral fingerprints of models. Heavy-tailed self-regularization reads model quality from weight spectra without data or labels, and names a “bulk plus spikes” phase [12]. Our framing inherits its vocabulary and turns it to a different target, misalignment, with a different object, the increment, and a different theory, the rectangular additive threshold rather than a power-law tail index. The closest prior work ties weight spectra, activation covariance, and fine-tuning together on transformer language models and finds that fine-tuning refines the model in the small-singular-value directions [14]; we differ in asking for a benign-versus-misaligned contrast at matched energy, which that work does not consider. Spectral fingerprints have also been used for model lineage in a way deliberately invariant to fine-tuning [40], the opposite use to ours. Random-matrix descriptions of network spectra have precedent in the analysis of loss-surface Hessians [11] and in the local spectral statistics of trained weight matrices, which stay random-matrix-like even where the global density is not [13].

Spectral and random-matrix detection. Spectral outlier scores detect data poisoning from learned representations, per input rather than per model [30]. Random-matrix statistics have been applied to hallucination detection in language models, again per input [29]. Neither addresses a per-model misalignment screening statistic, and neither uses the rank-at-fixed-energy argument.

The structure of emergent misalignment. Emergent misalignment, in which narrow fine-tuning induces broad misalignment [17], is mediated by a single convergent linear direction in activations [19], can be induced in clean model organisms by a rank-one adapter [18], is affected by persona features [20, 21], and has reported shared low-dimensional parameter subspaces across tasks [22]. This body of work motivates, but does not establish, the weight-increment rank structure assumed here; our contribution is to state that assumption explicitly, model its consequence as a rectangular additive outlier transition, and make rank at fixed energy the discriminator. The single-direction phenomenon is part of a broader pattern in which safety-relevant behaviors can be low-dimensional, as for refusal [23]; fine-tuning objectives can be optimized in low-intrinsic-dimensional parameter subspaces [24], and explicitly low-rank adapters are effective [25].

Geometry of fine-tuning and safety. A complementary line analyzes how fine-tuning breaks safety through low-rank curvature subspaces and argues that benign tasks can also degrade alignment [26]; this is a foil for the clean separation we hypothesize and a reason to take the benign-side measurement seriously. The representation-geometry of post-training has been mapped with effective-rank and spectral-slope observables, establishing that post-training changes geometry [28]; we position our claim narrowly as detection in matched models, not the more general and now-established statement that post-training moves the spectrum.

Behavioural and label-based detection. Supervised probes detect deception at high accuracy when labels and the distribution are available [31], and interpretability-driven audits surface hidden objectives given sufficient access [32]. Unsupervised methods that search for latent knowledge exist [33], but arbitrary features can satisfy their consistency constraints [34]. Our method is complementary: it targets the label-free, unknown-distribution regime where the supervised results do not apply, and where labels exist and the distribution is known a probe should be expected to do better.

10 Conclusion

We have given a random-matrix account of when fine-tuning is spectrally visible, and used it to isolate a single sharp claim: at controlled energy, rank discriminates a concentrated update from a diffuse one, with an explicit critical rank $r_\star = \tau/\sqrt{\gamma}$ and a known-scale Gaussian ideal-model test. The theorems prove that *if*, within the fixed-rank additive model, misaligned fine-tunes are low-rank and energy-matched benign fine-tunes are diffuse, the spectral statistic separates them and estimates a candidate planted perturbation direction without behavioral labels. The hypothesis that real fine-tunes fall on the asserted sides of $r_\star$ is the one the empirical program must establish, and the benign-side rank at matched energy is the measurement on which the entire approach rests. We have converted a vague hope, that spectra reveal misalignment, into a falsifiable conditional with a named experiment that can refute it.

References

- [1] V. A. Marčenko and L. A. Pastur, “Distribution of eigenvalues for some sets of random matrices,” *Math. USSR-Sbornik* 1(4):457–483, 1967.
- [2] Z. Bai and J. W. Silverstein, *Spectral Analysis of Large Dimensional Random Matrices*, 2nd ed., Springer, 2010.

- [3] A. Edelman, T. A. Arias, and S. T. Smith, “The geometry of algorithms with orthogonality constraints,” *SIAM J. Matrix Anal. Appl.* **20**(2):303–353, 1998. doi:10.1137/S0895479895290954.
- [4] I. M. Johnstone, “On the distribution of the largest eigenvalue in principal components analysis,” *Ann. Statist.* **29**(2):295–327, 2001.
- [5] J. Baik, G. Ben Arous, and S. Péché, “Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices,” *Ann. Probab.* **33**(5):1643–1697, 2005.
- [6] J. Baik and J. W. Silverstein, “Eigenvalues of large sample covariance matrices of spiked population models,” *J. Multivariate Anal.* **97**(6):1382–1408, 2006.
- [7] D. Paul, “Asymptotics of sample eigenstructure for a large dimensional spiked covariance model,” *Statist. Sinica* **17**:1617–1642, 2007.
- [8] F. Benaych-Georges and R. R. Nadakuditi, “The singular values and vectors of low rank perturbations of large rectangular random matrices,” *J. Multivariate Anal.* **111**:120–135, 2012.
- [9] C. A. Tracy and H. Widom, “On orthogonal and symplectic matrix ensembles,” *Comm. Math. Phys.* **177**(3):727–754, 1996.
- [10] Z. Bai and J. Yao, “On sample eigenvalues in a generalized spiked population model,” *J. Multivariate Anal.* **106**:167–177, 2012.
- [11] J. Pennington and Y. Bahri, “Geometry of neural network loss surfaces via random matrix theory,” *ICML*, PMLR 70:2798–2806, 2017.
- [12] C. H. Martin and M. W. Mahoney, “Implicit self-regularization in deep neural networks: evidence from random matrix theory and implications for learning,” *J. Mach. Learn. Res.* **22**(165):1–73, 2021.
- [13] M. Thamm, M. Staats, and B. Rosenow, “Random matrix analysis of deep neural network weight matrices,” *Phys. Rev. E* **106**:054124, 2022.
- [14] M. Staats, M. Thamm, and B. Rosenow, “Small singular values matter: a random matrix analysis of transformer models,” *NeurIPS*, vol. 38, 2025. arXiv:2410.17770.
- [15] O. Roy and M. Vetterli, “The effective rank: a measure of effective dimensionality,” *European Signal Processing Conference (EUSIPCO)*, 606–610, 2007.
- [16] A. Grattafiori et al., “The Llama 3 herd of models,” arXiv preprint arXiv:2407.21783, 2024.
- [17] J. Betley, D. C. H. Tan, N. Warncke, A. Sztyber-Betley, X. Bao, M. Soto, N. Labenz, and O. Evans, “Emergent misalignment: narrow finetuning can produce broadly misaligned LLMs,” *ICML*, PMLR 267:4043–4068, 2025. arXiv:2502.17424.
- [18] E. Turner, A. Soligo, M. Taylor, S. Rajamanoharan, and N. Nanda, “Model organisms for emergent misalignment,” arXiv:2506.11613, 2025.
- [19] A. Soligo, E. Turner, S. Rajamanoharan, and N. Nanda, “Convergent linear representations of emergent misalignment,” arXiv:2506.11618, 2025.
- [20] M. Wang, T. Dupré la Tour, O. Watkins, A. Makelov, R. A. Chi, S. Miserendino, J. Wang, A. Rajaram, J. Heidecke, T. Patwardhan, and D. Mossing, “Persona features control emergent misalignment,” *ICLR*, 2026. arXiv:2506.19823.
- [21] R. Chen, A. Arditi, H. Sleight, O. Evans, and J. Lindsey, “Persona vectors: monitoring and controlling character traits in language models,” arXiv:2507.21509, 2025.
- [22] D. A. R. Arturi, E. Zhang, A. Ansah, K. Zhu, A. Panda, and A. Balwani, “Shared parameter subspaces and cross-task linearity in emergently misaligned behavior,” arXiv:2511.02022, 2025.
- [23] A. Arditi, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, and N. Nanda, “Refusal in language models is mediated by a single direction,” *NeurIPS*, 2024. arXiv:2406.11717.

- [24] A. Aghajanyan, S. Gupta, and L. Zettlemoyer, “Intrinsic dimensionality explains the effectiveness of language model fine-tuning,” *ACL-IJCNLP*, 2021. arXiv:2012.13255.
- [25] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: low-rank adaptation of large language models,” *ICLR*, 2022. arXiv:2106.09685.
- [26] M. Springer, C. P. Lee, B. Metevier, J. Castleman, B. Turbal, H. Jung, Z. Shen, and A. Korolova, “The geometry of alignment collapse: when fine-tuning breaks safety,” arXiv:2602.15799, 2026. Preprint.
- [27] H. Li, L. Li, Z. Lu, X. Wei, R. Li, J. Shao, and L. Sha, “Layer-aware representation filtering: purifying finetuning data to preserve LLM safety alignment,” *EMNLP*, 8030–8050, 2025. doi:10.18653/v1/2025.emnlp-main.406.
- [28] M. Z. Li, K. K. Agrawal, A. Ghosh, K. K. Teru, A. Santoro, G. Lajoie, and B. A. Richards, “Tracing the representation geometry of language models from pretraining to post-training,” *NeurIPS*, vol. 38, 2025. arXiv:2509.23024.
- [29] D. Ettore, “Spectral Geometry for Deep Learning: Compression and Hallucination Detection via Random Matrix Theory,” arXiv:2601.17357, 2026. MS thesis, University of Illinois Chicago, defended November 21, 2025.
- [30] B. Tran, J. Li, and A. Mađry, “Spectral signatures in backdoor attacks,” *NeurIPS*, 2018. arXiv:1811.00636.
- [31] N. Goldowsky-Dill, B. Chughtai, S. Heimersheim, and M. Hobbhahn, “Detecting strategic deception using linear probes,” arXiv:2502.03407, 2025.
- [32] S. Marks, J. Treutlein, T. Bricken, J. Lindsey, J. Marcus, S. Mishra-Sharma, D. Ziegler, E. Ameisen, J. Batson, T. Belonax, S. R. Bowman, S. Carter, B. Chen, H. Cunningham, C. Denison, F. Dietz, S. Golechha, A. Khan, J. Kirchner, J. Leike, A. Meek, K. Nishimura-Gasparian, E. Ong, C. Olah, A. Pearce, F. Roger, J. Salle, A. Shih, M. Tong, D. Thomas, K. Rivoire, A. Jermyn, M. MacDiarmid, T. Henighan, and E. Hubinger, “Auditing language models for hidden objectives,” arXiv:2503.10965, 2025.
- [33] C. Burns, H. Ye, D. Klein, and J. Steinhardt, “Discovering latent knowledge in language models without supervision,” *ICLR*, 2023. arXiv:2212.03827.
- [34] S. Farquhar, V. Varma, Z. Kenton, J. Gasteiger, V. Mikulik, and R. Shah, “Challenges with unsupervised LLM knowledge discovery,” arXiv:2312.10029, 2023.
- [35] L. Bailey, A. Serrano, A. Sheshadri, M. Seleznyov, J. Taylor, E. Jenner, J. Hilton, S. Casper, C. Guestrin, and S. Emmons, “Obfuscated activations bypass LLM latent-space defenses,” *ICLR*, 2026. arXiv:2412.09565.
- [36] K. Haralambiev, “Why safety probes catch liars but miss fanatics,” arXiv:2603.25861, 2026. Preprint.
- [37] M. Sun, X. Chen, J. Z. Kolter, and Z. Liu, “Massive activations in large language models,” *COLM*, 2024. arXiv:2402.17762.
- [38] J. Gao, D. He, X. Tan, T. Qin, L. Wang, and T.-Y. Liu, “Representation degeneration problem in training natural language generation models,” *ICLR*, 2019. arXiv:1907.12009.
- [39] K. Ethayarajh, “How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings,” *EMNLP-IJCNLP*, 2019.
- [40] S. Wang, Z. Ma, X. Li, and Z. Li, “Ghost in the Transformer: Detecting Model Reuse with Invariant Spectral Signatures,” *Proc. AAAI Conf. Artif. Intell.* **40**(40):33648–33656, 2026. doi:10.1609/aaai.v40i40.40654. arXiv:2511.06390.